

HW 12

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Quiz 1

Algorithm quiz1(n)

int r1

if (n == 1)

r1 = 12

else

r1 = n + quiz1(n/2)

end if

return (r1)

end quiz1

quiz1(32) = ? 84

$$\text{quiz1}(32) = 32 + \text{quiz1}(16) = 32 + 52 = 84$$

$$\text{quiz1}(16) = 16 + \text{quiz1}(8) = 16 + 36 = 52$$

$$\text{quiz1}(8) = 8 + \text{quiz1}(4) = 8 + 28 = 36$$

$$\text{quiz1}(4) = 4 + \text{quiz1}(2) = 4 + 12 = 16$$

$$\text{quiz1}(2) = 2 + \text{quiz1}(1) = 2 + 12 = 14$$

$$\text{quiz1}(1) = 12$$

Quiz 2

Algorithm quiz2(n)

if (n==8)

println("Go Back!")

else

println(n + " Hello!")

quiz2(n+2)

println(n + " Bye!")

end if

end quiz2

quiz2(0) = ?

Quiz 3

Algorithm quiz3(x)

if (x<5)

return (3*x)

else

return (2*quiz3(x-5)+7)

end if

println("Finish!!")

return 1

end quiz3

quiz3(4) = ?

quiz3(10) = ?

quiz3(17) = ?

Algorithm quiz2(n)

if (n==8)

println("Go Back!")

else

println(n + " Hello!")

quiz2(n+2)

println(n + " Bye!")

end if

end quiz2

quiz2(0) = 0 Hello!
quiz2(2) = 2 Hello!
quiz2(4) = 4 Hello!
quiz2(6) = 6 Hello!
quiz2(8) = Go Back!
6 Bye!
4 Bye!
2 Bye!
0 Bye!

Algorithm quiz3(x)

if (x<5)

return (3*x)

else

return (2*quiz3(x-5)+7)

end if

println("Finish!!")

return 1

end quiz3

q 10 17

$$\text{quiz3}(4) = 3 * 4 = 12$$

$$\text{quiz3}(10) = 2 * \text{quiz3}(5) + 7 = 2 * 7 + 7 = 21$$

$$\text{quiz3}(5) = 2 * \text{quiz3}(0) + 7 = 2 * 0 + 7 = 7$$

$$\text{quiz3}(0) = 3 * 0 = 0$$

$$\text{quiz3}(17) = 2 * \text{quiz3}(12) + 7 = 2 * 45 + 7 = 97$$

$$\text{quiz3}(12) = 2 * \text{quiz3}(7) + 7 = 2 * 19 + 7 = 45$$

$$\text{quiz3}(7) = 2 * \text{quiz3}(2) + 7 = 2 * 6 + 7 = 19$$

$$\text{quiz3}(2) = 3 * 2 = 6$$